Singular points and Frobenius' Theorem

We already know from our discussion so far about power series solutions that a singular point for the differential equation

$$p(x)y'' + q(x)y' + r(x)y = 0$$

occurs wherever $p(x) = 0$. But singular points can actually be classified into two groups: regular and irregular.

A **regular singular point** occurs when both of these expressions are analytic at x_0:

$$Q(x) = (x - x_0)\frac{q(x)}{p(x)}$$

$$R(x) = (x - x_0)^2\frac{r(x)}{p(x)}$$

Otherwise, if one or both of these expressions are not analytic at x_0, then the singular point is an **irregular singular point**.

Frobenius' Theorem

To find a series solution to a differential equation around a regular singular point, we can use **Frobenius' Theorem**, which says that there is at least one solution to $p(x)y'' + q(x)y' + r(x)y = 0$ of the form

$$y = (x - x_0)^r \sum_{n=0}^{\infty} c_n(x - x_0)^n = \sum_{n=0}^{\infty} c_n(x - x_0)^{n+r}$$

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where r is a constant that we'll need to find. If the regular singular point is $x_0 = 0$, then of course this solution equation simplifies to

$$y = x^r \sum_{n=0}^{\infty} c_n x^n = \sum_{n=0}^{\infty} c_n x^{n+r}$$

So the **method of Frobenius** will have us substituting this solution equation and its derivatives

$$y = \sum_{n=0}^{\infty} c_n x^{n+r}$$

$$y' = \sum_{n=0}^{\infty} c_n (n+r) x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} c_n (n+r)(n+r-1) x^{n+r-2}$$

into the differential equation, and then simplifying and equating coefficients to find the **indicial equation**. The indicial equation will be a quadratic equation, which means that the **indicial roots** will be its solutions r_1 and r_2. There are three relationships between these roots that we want to consider.

Case I: When $r_1 > r_2$ and $r_1 - r_2$ is not a positive integer, then the differential equation has two linearly independent solutions,

$$y_1(x) = \sum_{n=0}^{\infty} c_n x^{n+r_1} \text{ with } c_0 \neq 0$$

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$$y_2(x) = \sum_{n=0}^{\infty} b_n x^{n+r_2} \text{ with } b_0 \neq 0$$

Case II: When $r_1 > r_2$ and $r_1 - r_2$ is a positive integer, then the differential equation has two linearly independent solutions,

$$y_1(x) = \sum_{n=0}^{\infty} c_n x^{n+r_1} \text{ with } c_0 \neq 0$$

$$y_2(x) = Cy_1(x)\ln x + \sum_{n=0}^{\infty} b_n x^{n+r_2} \text{ with } b_0 \neq 0$$

It's possible in this case that $C = 0$, which would zero out the first term, and the second solution $y_2(x)$ would not contain a logarithm.

Case III: But when $r_1 = r_2$, the second solution does not include the constant C, and therefore the second solution will definitely contain the logarithm.

$$y_1(x) = \sum_{n=0}^{\infty} c_n x^{n+r_1} \text{ with } c_0 \neq 0$$

$$y_2(x) = y_1(x)\ln x + \sum_{n=0}^{\infty} b_n x^{n+r_1} \text{ with } b_0 \neq 0$$

Let's do an example where we use the method of Frobenius to try to find a series solution to the differential equation around a regular singular point. This will be a Case I example, which means we should expect to find two linearly independent solutions that we can sum to find the general solution.

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Example

Determine the regularity of the singular point $x_0 = 0$ of the differential equation, use the method of Frobenius to build any solution(s) around that point, then find the general solution.

$$2xy'' - (3 + 2x)y' + y = 0$$

Matching this differential equation to the standard form $p(x)y'' + q(x)y' + r(x)y = 0$, we can identify

$$p(x) = 2x$$

$$q(x) = -(3 + 2x)$$

$$r(x) = 1$$

Using these three functions to calculate $Q(x)$ and $R(x)$ gives

$$Q(x) = (x - x_0)\frac{q(x)}{p(x)} = (x)\frac{-(3 + 2x)}{2x} = -\frac{3 + 2x}{2}$$

$$R(x) = (x - x_0)^2\frac{r(x)}{p(x)} = x^2\frac{1}{2x} = \frac{x}{2}$$

Because both denominators simplify to constants, we can see that both $Q(x)$ and $R(x)$ are analytic about $x_0 = 0$, so $x_0 = 0$ is a regular singular point.

Now we'll use

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$$y = \sum_{n=0}^{\infty} c_n x^{n+r}$$

$$y' = \sum_{n=0}^{\infty} (n+r)c_n x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1)c_n x^{n+r-2}$$

to make substitutions into the differential equation.

$$2xy'' - (3+2x)y' + y = 0$$

$$2x\sum_{n=0}^{\infty} (n+r)(n+r-1)c_n x^{n+r-2} - (3+2x)\sum_{n=0}^{\infty} (n+r)c_n x^{n+r-1} + \sum_{n=0}^{\infty} c_n x^{n+r} = 0$$

$$\sum_{n=0}^{\infty} 2(n+r)(n+r-1)c_n x^{n+r-1} - 3\sum_{n=0}^{\infty} (n+r)c_n x^{n+r-1} - 2x\sum_{n=0}^{\infty} (n+r)c_n x^{n+r-1}$$

$$+ \sum_{n=0}^{\infty} c_n x^{n+r} = 0$$

$$\sum_{n=0}^{\infty} 2(n+r)(n+r-1)c_n x^{n+r-1} - \sum_{n=0}^{\infty} 3(n+r)c_n x^{n+r-1}$$

$$- \sum_{n=0}^{\infty} 2(n+r)c_n x^{n+r} + \sum_{n=0}^{\infty} c_n x^{n+r} = 0$$

Combine terms with equivalent powers of x.

$$\sum_{n=0}^{\infty} [2(n+r)(n+r-1)c_n - 3(n+r)c_n]x^{n+r-1}$$

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$$+\sum_{n=0}^{\infty}[c_n - 2(n+r)c_n]x^{n+r} = 0$$

$$\sum_{n=0}^{\infty}[(2(n+r-1)-3)(n+r)c_n]x^{n+r-1} + \sum_{n=0}^{\infty}[(1-2(n+r))c_n]x^{n+r} = 0$$

$$\sum_{n=0}^{\infty}[(2n+2r-2-3)(n+r)c_n]x^{n+r-1} + \sum_{n=0}^{\infty}[(1-2n-2r)c_n]x^{n+r} = 0$$

$$\sum_{n=0}^{\infty}(2n+2r-5)(n+r)c_n x^{n+r-1} + \sum_{n=0}^{\infty}(1-2n-2r)c_n x^{n+r} = 0$$

Factor x^r out of the left side of the equation.

$$x^r\left[\sum_{n=0}^{\infty}(2n+2r-5)(n+r)c_n x^{n-1} + \sum_{n=0}^{\infty}(1-2n-2r)c_n x^n\right] = 0$$

Now we need to make sure these two series are in phase. The first series starts with the x^{-1} term, while the second series starts with the x^0 term. So we'll pull the x^{-1} term out of the first series to put them in phase.

$$x^r\left[r(2r-5)c_0 x^{-1} + \sum_{n=1}^{\infty}(2n+2r-5)(n+r)c_n x^{n-1}\right.$$

$$\left.+\sum_{n=0}^{\infty}(1-2n-2r)c_n x^n\right] = 0$$

Now that the series are in phase, we'll substitute $k = n-1$ and $n = k+1$ into the first series, and $k = n$ into the second series.

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$$x^r\left[r(2r-5)c_0x^{-1}+\sum_{k=0}^{\infty}(2k+2r-3)(k+r+1)c_{k+1}x^k\right.$$

$$\left.-\sum_{k=0}^{\infty}(2k+2r-1)c_kx^k\right]=0$$

Now that the series are in phase with matching indices, combine them.

$$x^r\left[r(2r-5)c_0x^{-1}\right.$$

$$\left.+\sum_{k=0}^{\infty}[(2k+2r-3)(k+r+1)c_{k+1}-(2k+2r-1)c_k]x^k\right]=0$$

This equation gives

$$r(2r-5)c_0=0$$

$$(2k+2r-3)(k+r+1)c_{k+1}-(2k+2r-1)c_k=0 \qquad k=0,\ 1,\ 2,\ \ldots$$

or

$$r(2r-5)=0$$

$$c_{k+1}=\frac{(2k+2r-1)c_k}{(2k+2r-3)(k+r+1)} \qquad k=0,\ 1,\ 2,\ \ldots$$

The indicial equation gives us $r_1=0$ and $r_2=5/2$. Substituting these indicial roots into the recurrence relation, we get

For $r_1=0$ For $r_2=5/2$

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	$c_{k+1} = \frac{(2k-1)c_k}{(2k-3)(k+1)}$	$c_{k+1} = \frac{(k+2)c_k}{(k+1)(k+\frac{7}{2})}$
$k = 0$	$c_1 = \frac{1}{3}c_0$	$c_1 = \frac{2 \cdot 2}{7}c_0$
$k = 1$	$c_2 = -\frac{c_1}{2} = -\frac{1}{6}c_0$	$c_2 = \frac{(2)(3)c_1}{(2)(9)} = \frac{2^2 \cdot 3}{9 \cdot 7}c_0$
$k = 2$	$c_3 = c_2 = -\frac{1}{6}c_0$	$c_3 = \frac{(2)(4)c_2}{(3)(11)} = \frac{2^3 \cdot 4}{11 \cdot 9 \cdot 7}c_0$
...	...	...

Forming these coefficients into series around $x_0 = 0$ gives,

$$c_0(x-x_0)^0 + c_1(x-x_0)^1 + c_2(x-x_0)^2 + c_3(x-x_0)^3 + \dots$$

$$c_0(x-0)^0 + \frac{1}{3}c_0(x-0)^1 - \frac{1}{6}c_0(x-0)^2 - \frac{1}{6}c_0(x-0)^3 - \dots$$

$$c_0\left(1 + \frac{1}{3}x - \frac{1}{6}x^2 - \frac{1}{6}x^3 - \dots\right)$$

and

$$c_0(x-x_0)^0 + c_1(x-x_0)^1 + c_2(x-x_0)^2 + c_3(x-x_0)^3 + \dots$$

$$c_0(x-0)^0 + \frac{2 \cdot 2}{7}c_0(x-0)^1 + \frac{2^2 \cdot 3}{9 \cdot 7}c_0(x-0)^2 + \frac{2^3 \cdot 4}{11 \cdot 9 \cdot 7}c_0(x-0)^3 + \dots$$

$$c_0\left(1 + \frac{2 \cdot 2}{7}x + \frac{2^2 \cdot 3}{9 \cdot 7}x^2 + \frac{2^3 \cdot 4}{11 \cdot 9 \cdot 7}x^3 + \dots\right)$$

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so the series solutions are

$$y_1(x) = c_0 x^0 \left(1 + \frac{1}{3}x - \frac{1}{6}x^2 - \frac{1}{6}x^3 - \dots\right)$$

$$y_2(x) = c_0 x^{\frac{5}{2}} \left(1 + \frac{2 \cdot 2}{7}x + \frac{2^2 \cdot 3}{9 \cdot 7}x^2 + \frac{2^3 \cdot 4}{11 \cdot 9 \cdot 7}x^3 + \dots\right)$$

and the general solution on the interval $(0,\infty)$ is

$$y(x) = C_1 y_1(x) + C_2 y_2(x)$$

$$y(x) = C_1 c_0 \left(1 + \frac{1}{3}x - \frac{1}{6}x^2 - \frac{1}{6}x^3 - \dots\right)$$

$$+ C_2 c_0 x^{\frac{5}{2}} \left(1 + \frac{2 \cdot 2}{7}x + \frac{2^2 \cdot 3}{9 \cdot 7}x^2 + \frac{2^3 \cdot 4}{11 \cdot 9 \cdot 7}x^3 + \dots\right)$$

But $C_1 c_0$ and $C_2 c_0$ are constants, so we can simplify them and write the general solution as

$$y(x) = C_1 \left(1 + \frac{1}{3}x - \frac{1}{6}x^2 - \frac{1}{6}x^3 - \dots\right)$$

$$+ C_2 x^{\frac{5}{2}} \left(1 + \frac{2 \cdot 2}{7}x + \frac{2^2 \cdot 3}{9 \cdot 7}x^2 + \frac{2^3 \cdot 4}{11 \cdot 9 \cdot 7}x^3 + \dots\right)$$

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